

Course introduction

08 Sep 2025

Land acknowledgement

McMaster University (and this class) is on the traditional territories of the Mississauga and Haudenosaunee nations, and within the lands protected by the “Dish with One Spoon” wampum agreement. (**Why?** See “Beyond Territorial Acknowledgements”)

Course goals

Primary: To learn about the ecology and evolution of infectious diseases and parasites in general, both (1) the E&E of infectious diseases (how E&E principles apply to pathogens) and (2) the effect of pathogens on the E&E of their host organisms and communities.

Secondary: to practice a variety of skills including critical and creative thinking; critical reading of papers from the primary and secondary literature; scientific writing and review; web and literature searches for scientific information.

Check out Perry Jr. (1997).

Expectations

- attend and participate in class (on time); do the reading; ask questions (including submitting your discussion questions online and participating in class and online discussions); think critically and creatively. *Please let me know if you can't participate in class discussion (shyness, internet/remote-participation issues, etc.).*
- know basic ecological and evolutionary principles (from BIO 1M03, 2C03, 2F03, 3FF3) but more importantly **to ask questions, in or out of class, when you don't understand something**. There is no such thing as a stupid question. If you don't know something you're probably not the only one in class.
- behave responsibly in online and in-person settings
- do your own work, don't plagiarize, reference sources appropriately, credit group work appropriately: see honesty.html.
- Appropriate use of generative AI. We will discuss GenAI at greater length later, but roughly:
 - **always** cite GenAI use (see citation guide)
 - fixing spelling and grammar: OK, make sure it doesn't change the meaning
 - polishing writing: if you must, although it might not improve it
 - finding resources: OK, but check that the sources really exist and that facts are correct
 - coming up with ideas: **not OK**
 - writing from scratch: **not OK** (not even first drafts)

Assignments & grading

- See here

Readings and discussion

In addition to the textbook, we will read papers from the primary and secondary literature and discuss them in class every Friday (starting September 12). I hope to do reading-based interactions on Perusall, still working out the details.

You will need to submit *discussion questions* (DQs) by 6 PM the day before discussion (i.e., Thursday). DQs are ideally conceptual questions — something you found interesting, or wrong, or puzzling about the paper and would like to discuss in class — but can also be about specific terms or points in the paper that you didn't understand and would like clarified in class.

Recordings and notes

- I will post lecture notes beforehand; recordings will be available by request. Don't expect them to be comprehensive.
- If you prefer notes in another format, ask.

Topics

- Way too much interesting stuff:
 - how does the ecology (“distribution and abundance”) of parasites/ID work? Classic ID epidemiology, population dynamics. Variation in space and time.
 - how do parasites/ID evolve? Phylogenetics, population genetics; phenotypes, genotypes
 - how do parasites/ID affect the ecology of their hosts? Biocontrol, conservation biology, apparent competition, trophic cascades
 - how do parasites/ID affect the evolution of their hosts? Coevolution
- The important questions:
 - **How do we know?** Population genetics, phylogenetics, historical records, surveys, experiments, statistics, mathematical models, meta-analysis . . .
 - **So what?** Why are we bothering with the details of a certain fact or definition?

References

Perry Jr., William G. 1997. “Examsmanship and the Liberal Arts: An Epistemological Inquiry.” In *The Writer's Home Companion: An Anthology of the World's Best Writing Advice, from Keats to Kunitz*, edited by Joan Bolker, 234–50. New York: Henry Holt & Company. <https://web.archive.org/web/20031219180243/http://plantpath.unl.edu/llane/101/examsmanship.html>.